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The rodent model of impaired gastric motility induced by allyl isothiocyanate, a pungent ingredient of wasabi, to evaluate therapeutic agents for functional dyspepsia

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ABSTRACT

Functional dyspepsia (FD) is thought to be mainly based on gastric motility dysfunction and chronic hypersensitivity, yet FD animal models has been reported a few. We studied to establish the mouse model of impaired gastric motility induced by a pungent ingredient of wasabi allyl isothiocyanate (AITC), which is reliable to evaluate prokinetic agents. Male ddY mice were used. Gastric motility was measured by ¹³C-acetic acid breath test in conscious mice. AITC (80 mM) was given 60 min before the measurement of motility. Prokinetic agents including itopride (30, 100 mg/kg), mosapride (0.1–1 mg/kg), neostigmine (30 µg/kg), acotiamide (10–100 mg/kg), and daikenchuto (100–1000 mg/kg) were given 40 min before the measurement. AITC impaired gastric motility without mucosal damages, which reverted 24 h after AITC treatment. The decreased motility induced by AITC was restored by prokinetic agents such as itopride, mosapride, neostigmine, and acotiamide. In separate experiment, daikenchuto recovered the decreased motility induced by AITC, although daikenchuto had no effect on motility in normal condition. In conclusion, it is considered that the AITC-induced impaired gastric motility mouse model is useful to develop new prokinetic agents for treatment of FD, and to re-evaluate traditional Japanese herbal medicines.

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1. Introduction

Functional dyspepsia (FD) is thought to be based on dysfunctional gastric motility and chronic hypersensitivity. Gastric motility impairment leads to symptoms of FD, including postprandial fullness/early satiety, nausea/vomiting, and/or bloating.¹ Acotiamide, a prokinetic agent, was shown to relieve postprandial fullness associated with FD.^{2,3}

Various FD positive animal models of gastric motility dysfunction, as follows, have been reported: (1) models of drug

administration (such as the model of clonidine administration),^{4,5} (2) neonatal intervention models (such as iodoacetamide-induced gastritis and neonatal maternal separation models),^{6,7} and (3) the low-grade inflammation model (using normoglycemic BioBreeding diabetes-prone rats).⁸ Although animal models of gastric motility dysfunction play a critical role in evaluating new drugs for FD, it is unclear which models are most suitable (and reliable) for studying the pathology of FD.

It has been shown that allyl isothiocyanate (AITC, a TRPA1 agonist and common ingredient of wasabi), induces acute neurogenic/non-neurogenic inflammation and pain.^{9,10} We have studied the pharmacological action of AITC in the gastrointestinal tract; according to our observations, changes in gastric mucosal integrity and gastric function have been observed upon intra-gastric application of AITC. As a result, we discovered that AITC disrupts the tight junction (TJ) barrier of the gastric mucosa,¹¹

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induces gastric mucosal hyperemia, and increases vascular permeability in rats.¹² Consequently, these findings revealed that AITC induced acute mucosal inflammation in rats. Therefore, we considered AITC to be a useful reagent for developing animal models of inflammatory gastric motility dysfunction.

In this study, we established a mouse model of impaired gastric motility induced by AITC, a pungent ingredient of wasabi. The ¹³C-labeled acetic acid breath test can continuously measure mouse gastric motility under conscious and non-invasive conditions.^{1–3} Therefore, we investigated: (1) the effect of AITC on gastric motility in normal mice, (2) the effect of the prokinetic agents on mice with AITC-induced gastric motility dysfunction, and (3) the effects of a traditional Japanese herbal medicine, daikenchuto, on mice with gastric motility dysfunction induced by AITC.

2. Material and methods

2.1. Animals

Male ddY mice (6–9 weeks old) were allowed to fast overnight before starting experiments. Gastric motility was measured according to established methods.^{14,15} The animal experiments were conducted per the “Guidelines for Care and Use of Experimental Mice” approved by the Japanese Pharmacological Society and the guidelines approved by the Josai International University Animal Experiment Ethics Committee (# 86). The minimum number of mice was used to produce the desired effect size and intensity (4–19 mice per group).

2.2. Experimental procedures

After oral administration of ¹³C-labeled acetic acid (64 mg/kg), the mouse was placed in a chamber and pumped to aspirate ¹³CO₂ excreted from exhaled breath. The rate of exhalation was measured using breath bags every 5 min until 90 min after the administration of acetic acid. The amount of ¹³CO₂ in the breath was measured using a POcone infrared spectrophotometer (Otsuka Electronics Co., Ltd.). ¹³C-labeled acetic acid is excreted from the stomach by gastric motility, absorbed, metabolized, and exhaled as ¹³CO₂. The amount of ¹³CO₂ exhaled was measured by calculating the difference in the amount of ¹³C with standard gas (95% O₂ + 5% CO₂) to determine the change in gastric motility. The breath test is a promising method for measuring gastric emptying. But we recognized the limitation of this method. The pulmonary output is a long distance from gastric input, although the methodological relevance is based on a close correspondence between gastric emptying of [¹³C]-acetic acid and pulmonary excretion of [¹³CO₂]. It has been shown the full amount of the [¹³C] label was not recovered in the breath, even though 100% dose of the [¹³C] label was emptied from the stomach in human. That is because (1) [¹³C] label could be fixed in metabolic products such as glucose, amino acid, and cholesterol, and (2) [¹³C] label could also retain as bicarbonate in systemic blood circulation, being eliminated in not only breath but also urine, stool, and sweat.¹⁶ But, no change was observed in animal condition and excretion in this study. AITC (1–100 mM, 0.1 mL/10 g body weight) in distilled water containing 1% Tween-80 was administered 30 or 60 min before the administration of ¹³C-labeled acetic acid. Test drugs were administered via subcutaneous injection 40 min before ¹³C-labeled acetic acid administration. In a few cases, 100 mM AITC induced adverse effects in animals before concluding experiments. Therefore, we decided to reduce the concentration of AITC to 80 mM; this new concentration reproducibly decreased gastric motility without producing mortality. Prokinetic agents such as itopride (a dual dopamine D₂ receptor and cholinesterase inhibitor: 30, 100 mg/kg), mosapride (a serotonin 5-HT₄ receptor agonist: 0.3,

1 mg/kg), neostigmine (a cholinesterase inhibitor dissolved in 1.5% lactic acid: 30 µg/kg), and acotiamide (10–100 mg/kg) were subcutaneously administered 40 min before the start of measurement. Daikenchuto (100–3000 mg/kg) was orally administered 40 min before the start of measurement. The impaired motility was observed throughout 3 h after oral administration of AITC (80 mM) in this study. But it was found those impaired motilities were not appeared about 6 h after oral administration of AITC as a pilot study, and those were completely disappeared at 24 h after oral administration. Therefore, we made an experimental design that AITC (80 mM) was given to mice at 60 min before the measurement of gastric motility. That is because (1) to produce impaired gastric motility as a pathological condition, and (2) to evaluate the effects of prokinetic drugs including DKT after the on-setting impaired motility induced by AITC. In addition, it was reported the timing of the administration of prokinetic drugs such as mosapride, itopride, neostigmine, and acotiamide was referred to reports.^{17,18} We also examined whether the AITC-induced decrease in gastric motility was dependent upon damage to the gastric mucosa in mice. Therefore, after the administration of AITC, we examined the gastric mucosa using a digital camera.

2.3. Preparations and drugs used

The ¹³C-labeled acetic acid used in this experiment was purchased from Cambridge Isotope Laboratories, Inc. (Andover, MA, USA). AITC, itopride, mosapride, and neostigmine were purchased from Wako Pure Chemical Industries, Ltd. (Osaka). Acotiamide was purchased from Asta Tech, Inc. (Bristol, PA, USA). Daikenchuto extract powder (Lot No. 2170100010) and powdered syrup (Lot No. 3620047) were gifts from Tsumura & Co. Ltd. (Tokyo). Daikenchuto is prepared by mixing DKT extract powder and powdered syrup at a ratio of 1:8.

2.4. Statistical analysis

Sigma Stat 3.1 software (Jandel Scientific Software, San Rafael, CA, USA) was used for all statistical analyses. The data are presented as the mean ± S.E. Differences between two groups were assessed via Student's *t*-test. Multiple comparisons against a single control group were made via a one-way analysis of variance (ANOVA) with Dunnett's test with *p* < 0.05 denoting statistical significance. To evaluate the degree of gastric motility dysfunction and improvement, the ratio of *C*_{max} (percentage) = $\Delta C_{\text{max of prokinetic agents' increase}}$ (prokinetic agents in AITC-treated mice/no prokinetic agent in AITC-treated mice)/ $\Delta C_{\text{max of AITC-induced motility dysfunction}}$ (AITC-treated mice/normal mice) was calculated.

3. Results

3.1. Effect of AITC on the mouse gastric motility in ¹³C-breath tests

The $\Delta^{13}\text{CO}_2$ increased in a time-dependent manner, reaching a peak at $1272.3 \pm 39.5\%$ at 20 min in normal mice (Fig. 1A). The $\Delta^{13}\text{CO}_2$ gradually decreased to baseline levels after reaching a peak at the end of experimentation. To determine the optimal concentration and responsiveness to AITC, we investigated the effects of AITC concentration on the time course of $\Delta^{13}\text{CO}_2$ in expired air from mice. AITC (1–100 mM) slowed gastric motility and the AUC_{90min} decreased in a luminal concentration-dependent manner (Fig. 1A,B). At 30 and 100 mM of AITC, the AUC_{90min} showed a clear decrease compared to the lower concentrations of AITC (1–10 mM) (Fig. 1B). AITC decreased the *C*_{max} value and delayed the *T*_{max} value in a concentration manner; there was a significant decrease in *C*_{max} and delay in *T*_{max} after treatment with 100 mM AITC compared with

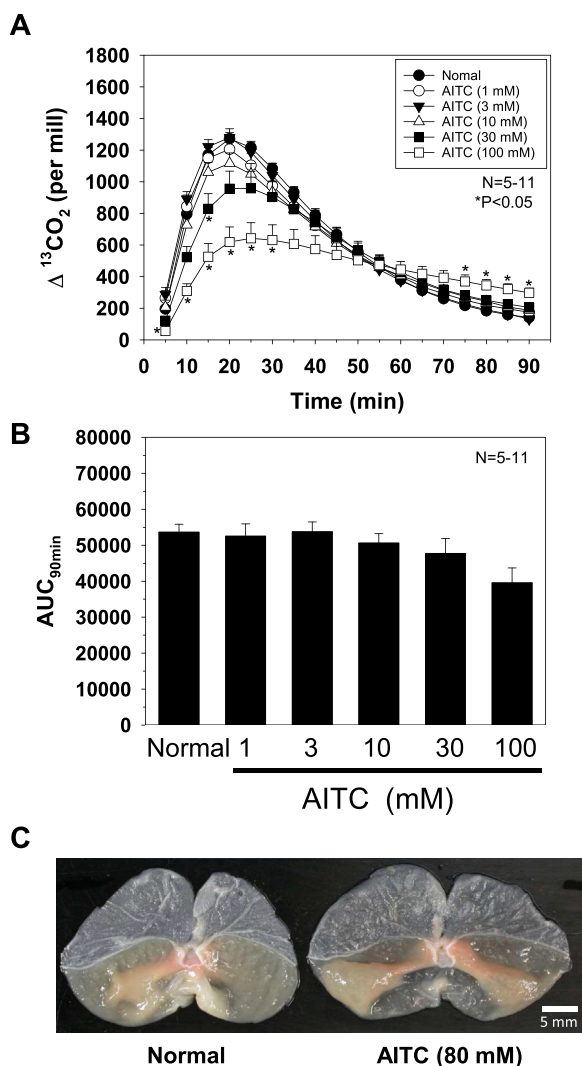


Fig. 1. Effect of allyl isothiocyanate (AITC) on the gastric emptying time course of $^{13}\text{CO}_2$ in expired air from conscious mice. **A**, The time course of $^{13}\text{CO}_2$ in expired air from conscious mice. AITC (1–100 mM) was administered intragastrically 40 min before the administration of ^{13}C -labeled acetic acid. The values represent the mean $\pm$ SE of 5–11 mice. **B**, The $\text{AUC}_{90\text{min}}$ of $^{13}\text{CO}_2$ in expired air from conscious mice. AITC (1–100 mM) was administered intragastrically 40 min before the administration of ^{13}C -labeled acetic acid. The values represent the mean $\pm$ SE of 5–11 mice. Gastric emptying was attenuated by AITC (1–100 mM) in a dose-dependent manner, and the dose of 100 mM of AITC prominently reduced gastric emptying until 50 min after the administration of ^{13}C -labeled acetic acid in conscious mice. The asterisk (*) denotes values that were significantly different from mice treated with vehicle by Dunnett's test (* $P < 0.05$). **C**, the typical appearance of the mouse gastric mucosa after the administration of AITC (80 mM) at the end of the experiments. Note that the hemorrhagic gastric lesion was not observed after the administration of AITC (80 mM) in conscious mice. Scale bar = 5 mm.

vehicle treatment (Supplemental Table 1). The writhing, hyper-spontaneous motor activity, and/or skeletal relaxation were observed in 2 mice when we gave 100 mM of AITC to 13 mice. It was noticed that AITC was oxidized easily. It was observed in a few cases by the administration of AITC which had been stored a long time, suggesting that degraded AITC was made by a long period and could induce writhing, skeletal muscle relaxation, and/or hyper-spontaneous motor activity as adverse effects. Thereafter, we replace one every 3–4 weeks to prevent degraded AITC from increasing adverse effects. We re-examined the concentration of AITC that decreased gastric motility. Therefore, the following experiments were carried out using 80 mM AITC to impair gastric

motility. No gastric mucosal damage was observed at the end of the experiments (Fig. 1C).

Gastric motility was decreased in mice treated with AITC (80 mM) 30 min before the administration of ^{13}C -labeled acetic acid compared to normal mice, resulting in a $619.6 \pm 77.4\%$ C_{max} at a T_{max} of 55.0 ± 11.7 min (Fig. 2A, B, and Supplemental Table 2). However, treatment with AITC (80 mM) did not suppress gastric motility after 24 h, thus resulting in a C_{max} of $1404.7 \pm 62.8\%$ at 19.0 ± 1.9 min T_{max} (Fig. 2C, D, and Supplemental Table 2). These data indicated that 80 mM AITC-induced gastric motility impairment was not maintained over 24 h and did not cause gastric mucosal damage.

3.2. Effects of prokinetic agents on AITC-attenuated gastric motility in mice

Next, we evaluated the effects of prokinetic agents, such as itopride, mosapride, and neostigmine, on AITC-induced gastric motility in mice. AITC (80 mM) was given 60 min before ^{13}C -labeled acetic acid administration in mice. Impaired gastric motility was observed after AITC administration, thus resulting in a C_{max} of $243.6 \pm 36.0\%$, and a C_{max} ratio of 41%. T_{max} was 25.0 ± 2.0 min in normal mice but was significantly delayed to 80.0 ± 8.4 min in AITC-treated mice (Fig. 3A and Table 1). Itopride (a dopamine D_2 receptor blocker that also inhibits cholinesterase activity) significantly recovered the gastric motility impairment induced by AITC. A C_{max} of $711.5 \pm 112.5\%$ was observed after treatment with 30 mg/kg of itopride, while a C_{max} of $1030.0 \pm 121.4\%$ was observed after treatment with 100 mg/kg of itopride. These results indicated that 30 mg/kg and 100 mg/kg itopride recovered impairment by about 46.0% and 77.3%, respectively. Itopride significantly recovered T_{max} and $\text{AUC}_{90\text{min}}$ in a dose-dependent manner (Table 1). Moreover, the AITC-induced impairment in gastric motility was significantly recovered by mosapride (1 mg/kg) treatment, resulting in a C_{max} restoration ratio of roughly 47.7% compared to AITC alone (Fig. 3B). Mosapride (1 mg/kg) also restored T_{max} (from 50.7 ± 10.6 min to 37.5 ± 8.4 min) and $\text{AUC}_{90\text{min}}$ (from $34,067.6 \pm 4624.5\%$ $\times$ min to $52,577.7 \pm 5468.1\%$ $\times$ min) in AITC-treated mice (Table 1). However, the gastric motility recovery (%) of AITC-treated mice was lower in response to mosapride than that to itopride.

The AITC-induced impairment in gastric motility was significantly recovered by neostigmine (30 $\mu\text{g/kg}$) treatment, resulting in a C_{max} restoration ratio of about 45.8% compared to AITC alone (Fig. 3C). Moreover, neostigmine (30 $\mu\text{g/kg}$) restored T_{max} (from 70.0 ± 10.2 min to 40.0 ± 13.4 min) and $\text{AUC}_{90\text{min}}$ (from $14,584.8 \pm 1354.2\%$ $\times$ min to $41,100.1 \pm 6660.7\%$ $\times$ min) in AITC-treated mice (Table 1). These results suggested that neostigmine (a cholinesterase inhibitor) recovered AITC-induced gastric motility impairment similar to itopride, which inhibits cholinesterases and antagonizes dopamine D_2 receptors.

3.3. Effect of the acotiamide on AITC-attenuated gastric motility in mice

Impaired gastric motility caused by AITC was recovered after treatment with acotiamide (10–100 mg/kg), resulting in a C_{max} restoration ratio of about 36.1% from acotiamide (100 mg/kg) compared to AITC alone (Fig. 4A). In addition, acotiamide (100 mg/kg) restored T_{max} (from 70.0 ± 10.2 min to 40.0 ± 13.4 min) and $\text{AUC}_{90\text{min}}$ (from $14,584.8 \pm 1354.2\%$ $\times$ min to $41,100.1 \pm 6660.7\%$ $\times$ min) in AITC-treated mice (Table 2 and Fig. 4B). These results suggested that acotiamide, a cholinesterase inhibitor and a therapeutic agent for FD, could recover AITC-induced decrease in gastric motility.

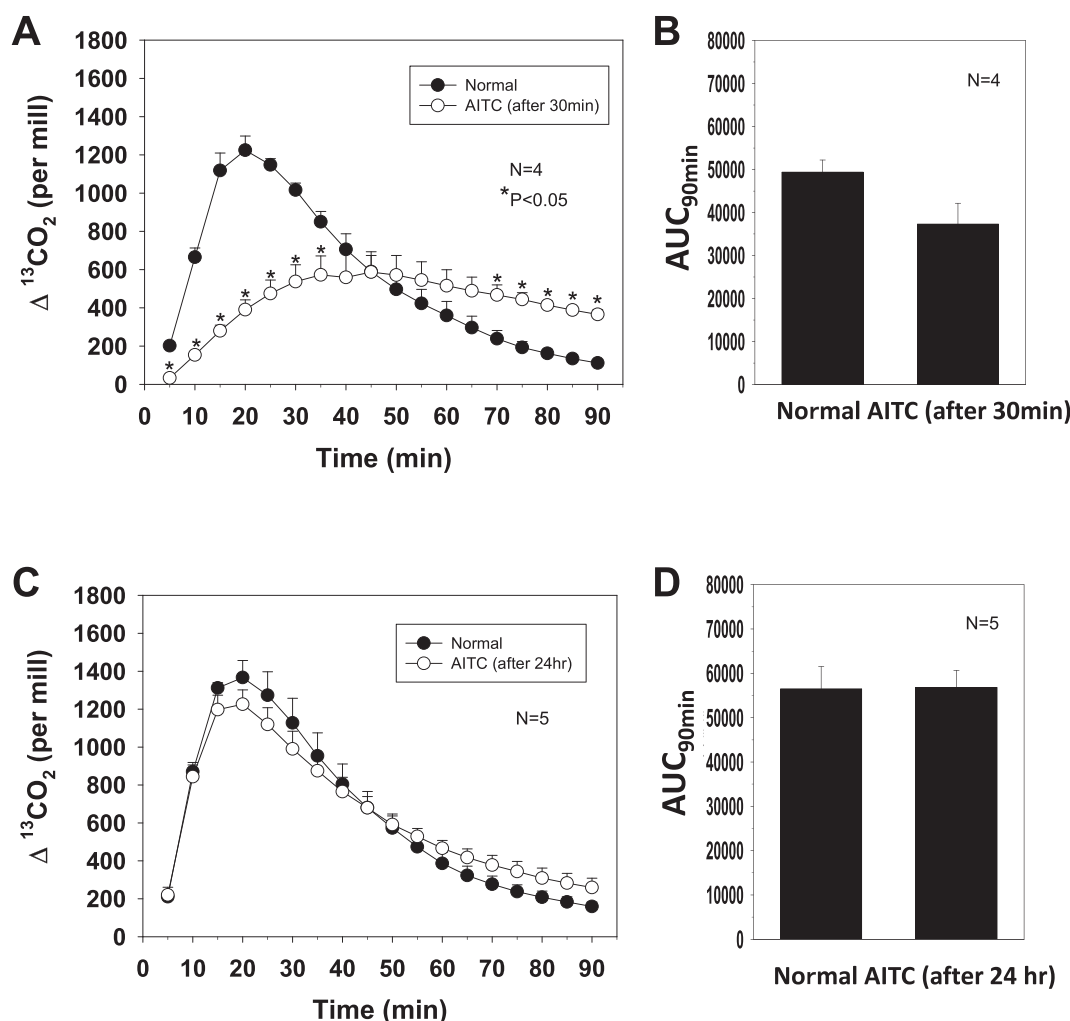


Fig. 2. The effect of AITC (80 mM) on gastric emptying at 30 min or 24 h after the administration of AITC in conscious mice. Gastric motility was completely inhibited after 30 min administration (Fig. 2A, B), but was not inhibited at all after 24 h (Fig. 2C, D). The asterisk (*) denotes values that were significantly different from normal mice as determined by Student's *t*-test (**P* < 0.05). **A**, The time course of $^{13}\text{CO}_2$ in expired air from conscious mice. AITC (80 mM) was administered intragastrically 30 min before the administration of ^{13}C -labeled acetic acid. The values represent the mean $\pm$ SE of 4 mice. **B**, $\text{AUC}_{90\text{min}}$ of $^{13}\text{CO}_2$ in expired air from conscious mice. AITC (80 mM) was administered intragastrically 30 min before the administration of ^{13}C -labeled acetic acid. The values represent the mean $\pm$ SE of 4 mice. **C**, The time course of $^{13}\text{CO}_2$ in expired air from conscious mice. AITC (80 mM) was administered intragastrically 24 h before the administration of ^{13}C -labeled acetic acid. The values represent the mean $\pm$ SE of 5 mice. **D**, $\text{AUC}_{90\text{min}}$ of $^{13}\text{CO}_2$ in expired air from conscious mice. AITC (80 mM) was administered intragastrically 24 h before the administration of ^{13}C -labeled acetic acid. The values represent the mean $\pm$ SE of 5 mice.

3.4. Effect of a traditional Japanese herbal medicine, daikenchuto, on gastric motility in mice

To evaluate the use of the AITC-induced model of gastric motility impairment for screening prokinetic agents, we subsequently examined the effect of daikenchuto on gastric motility. Daikenchuto (100–3000 mg/kg) did not increase gastric motility in normal mice (Fig. 5B and Table 3). However, daikenchuto (100–1000 mg/kg) increased gastric motility in AITC-treated mice in a dose-dependent manner, with a 1000 mg/kg dose significantly increasing gastric motility (Fig. 5A). Indeed, the C_{max} ratio was restored to about 51.5% by daikenchuto (1000 mg/kg) compared to AITC alone (Table 3). Additionally, a 3000 mg/kg dose of daikenchuto nullified the increase in gastric motility among AITC-treated mice of the smaller doses. While no changes in the T_{max} and $\text{AUC}_{90\text{min}}$ values were observed in response to daikenchuto (100–3000 mg/kg) in normal mice, daikenchuto (100–1000 mg/kg) dose-dependently restored T_{max} (from

45.6 ± 4.9 min to 26.4 ± 1.4 min) and $\text{AUC}_{90\text{min}}$ (from $36,146.6 \pm 211.2\% \times \text{min}$ to $48,719.7 \pm 3522.9\% \times \text{min}$) in mice with AITC-induced impaired gastric motility. However, 3000 mg/kg daikenchuto abolished the recovery of AITC-induced impairments in gastric motility.

4. Discussion

These results demonstrated that the oral administration of AITC markedly attenuated gastric motility in a concentration-dependent manner without producing gastric mucosal damage in mice. Furthermore, we showed that AITC-induced gastric motility impairment was attenuated after treatment with itopride (a dopamine D_2 receptor blocker/cholinesterase inhibitor), mosapride (a serotonin 5-HT $_4$ agonist), neostigmine (a cholinesterase inhibitor), and daikenchuto.

By establishing a mouse model of AITC-induced gastric motility impairment, we may better evaluate the efficacy of prokinetic

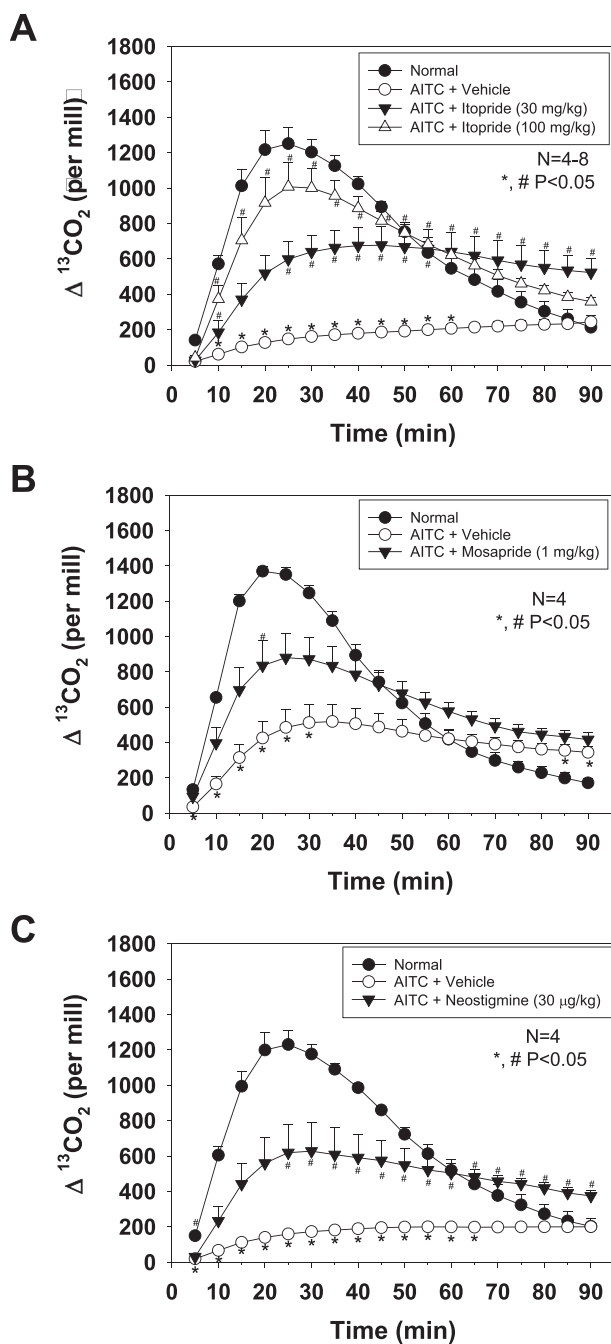


Fig. 3. A. Effect of itopride, a dual dopamine D₂ receptor and cholinesterase inhibitor, on the AITC-induced decrease in gastric motility in conscious mice. Itopride (30, 100 mg/kg, s.c.) was administered 20 min after administration of AITC. AITC (80 mM) was administered intragastrically 60 min before the administration of ¹³C-labeled acetic acid. Note that itopride significantly increased the AITC-impaired gastric motility in a dose-dependent manner. Each value represents the mean $\pm$ SE of 4–8 mice. The asterisk (*) denotes values that were significantly different from normal mice as determined by Student's *t*-test (**P* < 0.05). The hash mark (#) denotes values that were significantly different from mice treated with AITC + vehicle by Dunnett's test (#*P* < 0.05). B. The effect of mosapride, a serotonin 5-HT₄ receptor agonist, on AITC-induced decrease in gastric motility in conscious mice. Mosapride (1 mg/kg, s.c.) was administered 20 min after administration of AITC. AITC (80 mM) was administered intragastrically 60 min before the administration of ¹³C-labeled acetic acid. Each value represents the mean $\pm$ SE of 4 mice. The asterisk (*) denotes values that were significantly different from normal mice by Student's *t*-test (**P* < 0.05). The hash mark (#) denotes values that were significantly different from mice treated with AITC + vehicle as determined by Student's *t*-test (#*P* < 0.05). C. The effect of neostigmine, a cholinesterase inhibitor, on AITC-induced decrease in gastric motility in conscious mice. Neostigmine (30 μg/kg, s.c.) was administered 20 min after

agents and daikenchuto (which is known in traditional Japanese herbal medicine to facilitate gastrointestinal motility). Although daikenchuto does not affect the gastric motility of normal mice, it significantly enhanced gastric motility in AITC-treated mice.

Additionally, we studied the pharmacological actions of AITC (a pungent wasabi ingredient and TRPA1 activator) in the gastrointestinal tract to reveal the relationship between the functional roles of TRPA1 and the TRPV1 capsaicin receptors. Our group, as well as other independent groups, has reported the functional expression of TRPA1 in neuronal and non-neuronal cells of the gastrointestinal tract.^{11,19,20} These studies showed that AITC induced the collapse of gastric epithelial tight junction (TJ) barriers independent of the TRPA1 blocker ruthenium red in primary cultured rat gastric epithelial cells. These results indicated that AITC induced the breakdown of the gastric epithelial TJ barrier independent of TRPA1¹¹. Moreover, in ex-vivo studies, AITC increased gastric mucosal blood flow and vascular permeability in a concentration-dependent manner in the rat stomach. Interestingly, no gastric mucosal damage in response to AITC (10–100 mM) was found macroscopically at the end of the experiment. Increase in gastric mucosal blood flow, but not in vascular permeability, was inhibited by TRPA1 antagonism. It was reported that AITC-induced gastric mucosal inflammation as evidenced by increased mucosal blood flow (TRPA1-dependent) and vascular permeability (TRPA1-independent) without causing mucosal damage.¹² These results suggest that AITC impaired gastric motility by increasing mucosal inflammation in mice. Indeed, it was observed that the application of AITC to the rat stomach disrupted the mucosal TJ barrier and subsequently increased acid back-diffusion. This acid back-diffusion increase enhanced gastric mucosal blood flow and vascular permeability without causing gastric damage.²¹ It was shown that patients with FD suffer from gastroduodenal symptoms without evidence of a structural disease such as mucosal damage endoscopically.²² And the recent research indicated that impaired gastroduodenal mucosal integrity, which is no difference microscopically, are produced in patients with FD by attenuated mucosal tight junctional barrier.²³ On the other hand, it is well known that increased acid back diffusion after gastric mucosal barrier disruption induced by bile acid facilitates the increased mucosal blood flow to eliminate acid and to prevent ulceration.²⁴ Previously, we investigated the effect of AITC on acid back-diffusion and ulcerogenesis in the rat stomachs. It was observed that mucosal application of AITC (30–300 mM) significantly increased acid back diffusion and mucosal blood flow in the concentration-dependent manner, but no gastric damage even at AITC (300 mM) was macroscopically observed at the end of the experiment. However, it was found that increased acid back diffusion in response to AITC (300 mM) developed hemorrhagic mucosal damage in rats when mucosal blood flow increase in response to AITC was attenuated.¹² Therefore, it is considered that mucosal TJ barrier disruption in the development of FD facilitate to increase acid back diffusion, which may be followed by the impaired gastroduodenal integrity and low-grade inflammation. However, the acid back diffusion also increased mucosal blood flow, resulting in no mucosal damage microscopically.

Conversely, it has been shown that gastric motility was attenuated by AITC via TRPA1 activation in rats, suggesting that TRPA1 activation with AITC and methyl syringate decreased gastric

administration of AITC. AITC (80 mM) was administered intragastrically 60 min before the administration of ¹³C-labeled acetic acid. Note that Neostigmine significantly recovered the AITC-impaired gastric motility. Each value represents the mean $\pm$ SE of 4 mice. The asterisk (*) denotes values that were significantly different from normal mice by Student's *t*-test (**P* < 0.05). The hash mark (#) denotes values that were significantly different from AITC + vehicle by Student's *t*-test (#*P* < 0.05).

Table 1
Effect of prokinetic agents including itopride, mosapride, neostigmine on gastric motility attenuated by AITC in conscious mice.

agent	dose (mg/kg)	n	Cmax (%)	Tmax (min)	AUC (% × min)
Itopride	Normal	4	1260.5 ± 89.2	25.0 ± 2.0	61,429.8 ± 2813.1
	Vehicle	4	243.6 ± 36.0 *	80.0 ± 8.4 *	14,872.6 ± 1390.7 *
	30	4	711.5 ± 112.5 [#]	40.0 ± 6.5 [#]	46,871.0 ± 6606.5 [#]
	100	4	1030 ± 12.4 [#]	27.5 ± 2.5 [#]	56,228.8 ± 4768.1 [#]
Mosapride	Normal	4	1385.5 ± 29.6	23.8 ± 1.3	58,196.0 ± 2242.0
	Vehicle	7	564.6 ± 87.2 *	50.7 ± 10.6	34,067.6 ± 4624.5 *
	1	8	956.1 ± 95.0 [#]	37.5 ± 8.4	52,577.7 ± 5468.1 [#]
Neostigmine	Normal	4	1239.7 ± 76.9	23.8 ± 2.4	59,430.9 ± 1047.2
	Vehicle	4	212.6 ± 19.6 *	70.0 ± 10.2*	14,584.8 ± 1354.2 *
	30 (μg/kg)	4	683.1 ± 12.2 [#]	40.0 ± 13.4	41,100.1 ± 6660.7 [#]

*P < 0.05 vs normal.

[#]P < 0.05 vs vehicle.

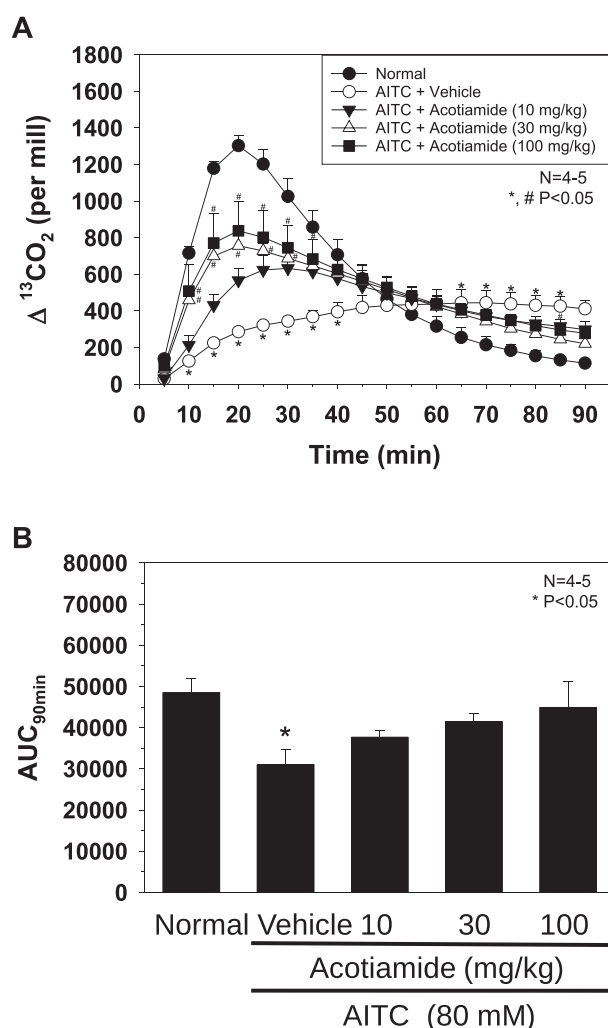


Fig. 4. Effect of acotiamide, a cholinesterase inhibitor used to treat functional dyspepsia, on AITC-induced decrease in gastric motility in conscious mice. Acotiamide (10–100 mg/kg) was administered 20 min after administration of AITC. AITC (80 mM) was administered intragastrically 60 min before the administration of ¹³C-labeled acetic acid. Note that acotiamide increased the AITC-impaired gastric motility in a dose-dependent manner. Each value represents the mean ± SE of 4–5 mice. The asterisk (*) denotes values that were significantly different from normal mice according to Student's *t*-test (*P < 0.05). The hash mark (#) denotes values that were significantly different from mice treated with AITC + vehicle by Dunnett's test (#P < 0.05). **A**, The time course of ¹³CO₂ in expired air from conscious mice. **B**, AUC_{90min} of ¹³CO₂ in expired air from conscious mice.

motility.^{25,26} However, in our preliminary study, AITC-induced decrease in gastric motility was not restored by a TRPA1 channel blocker A967079 (unpublished observation). Our results were consistent with those of the study by Capasso et al., in which gastric motility was attenuated by AITC (i.p. or p.o.) in a TRPA1-independent manner.²⁷

We clearly showed that AITC decreased gastric motility in mice, and itopride significantly restored motility in a concentration-dependent manner. Additionally, mosapride recovered the AITC-induced attenuation in gastric motility. This data suggests that the recovery of impaired gastric motility is attributable to increased parasympathetic activity and endogenous ACh release following treatment with itopride and mosapride.^{28,29} Furthermore, we studied the effect of neostigmine on AITC-induced impairments in gastric motility. Neostigmine caused the accumulation of endogenous ACh around gastrointestinal smooth muscle²⁹ and induced recovery. These results suggest that cholinesterase inhibition also contributed to the recovery from AITC-induced gastrointestinal dysfunction. The ability of itopride to inhibit cholinesterase activity and dopamine D₂ receptors could be important for recovering gastric motility. Moreover, AITC-induced gastric motility was not observed at 24 h after AITC administration, which suggests that this attenuation was reversible. Therefore, AITC may induce temporary desensitization of parasympathetic nerve endings in the gastrointestinal tract. Based on these results, this reproducible model of AITC-induced gastric motility impairment is sufficient to evaluate prokinetic agents for treating FD.

The ¹³C-labeled acetic acid breath test is used to continuously measure mouse gastric motility under non-invasive and conscious conditions.^{13–15} This method differs from the conventional phenol red method, which only examines endpoints. However, it is difficult to measure the increased motility in normal mice treated with itopride and mosapride using the ¹³C-labeled acetic acid breath test. Since traditional Japanese herbal medicines are less likely to show *in vivo* pharmacological effects compared to prokinetic agents depending upon the experimental conditions, it was necessary to

Table 2
Effect of the therapeutic agent for functional dyspepsia acotiamide on gastric motility attenuated by AITC in conscious mice.

agent	dose (mg/kg)	n	Cmax (%)	Tmax (min)
Acotiamide	Normal	4	1313.2 ± 48.3	19.0 ± 1.3
	Vehicle	4	462.1 ± 73.8 *	68.8 ± 7.7 *
	10	4	661.7 ± 78.6	32.5 ± 7.8 [#]
	30	4	790.2 ± 37.0	21.7 ± 4.4 [#]
	100	4	845.0 ± 162.0	20.0 ± 2.0 [#]

*P < 0.05 vs normal.

[#]P < 0.05 vs vehicle.

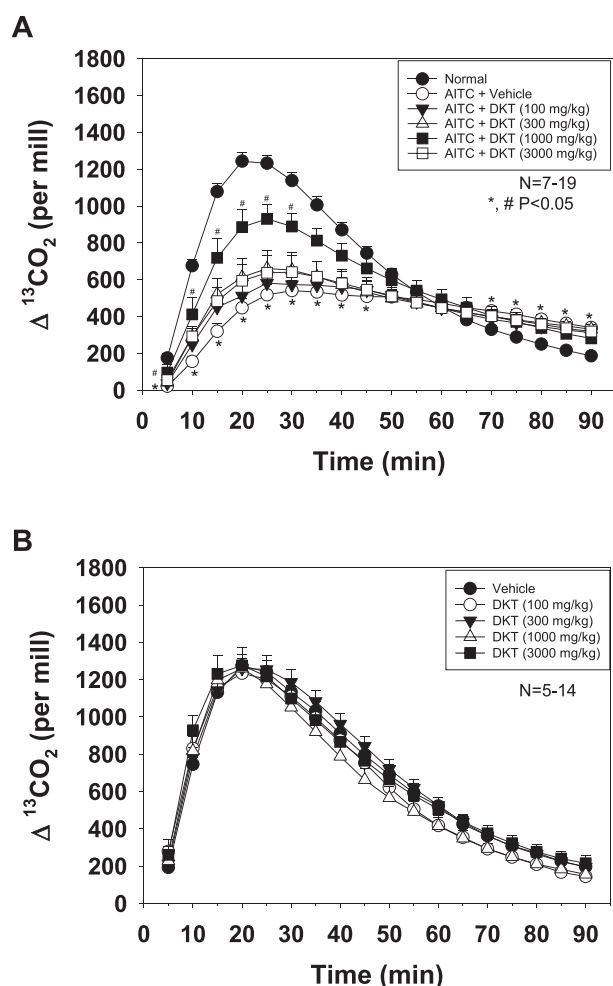


Fig. 5. Effect of daikenchuto (DKT) on AITC-induced decrease in gastric motility in conscious mice. **A**, The time course of $^{13}\text{CO}_2$ in expired air from mice that were pretreated with AITC. DKT (100–3000 mg/kg, p.o.) was administered 20 min after administration of AITC. AITC (80 mM) was administered intragastrically 60 min before the administration of ^{13}C -labeled acetic acid. Each value represents the mean $\pm$ SE of 7–19 mice. The asterisk (*) denotes values that were significantly different from mice with normal mice according to Student's *t*-test (**P* < 0.05). The hash mark (#) denotes values that were significantly different from mice treated with AITC + vehicle by Dunnett's test (#*P* < 0.05). **B**, The time course of $^{13}\text{CO}_2$ in expired air from normal mice. DKT (100–3000 mg/kg, p.o.) was administered 40 min before the experiment. Each value represents the mean $\pm$ SE of 5–14 mice. Note that DKT (100–3000 mg/kg) failed to increase gastric motility in normal mice (A). However, DKT (1000 mg/kg) significantly recovered gastric motility attenuated by AITC in mice.

Table 3
Effect of daikenchuto (DKT) on gastric motility attenuated by AITC in conscious mice.

agent	Dose (mg/kg)	n	Cmax (%)	Tmax (min)
(a) DKT AITC-treated mice	Normal	19	1279.2 $\pm$ 45.5	21.8 $\pm$ 1.0
	Vehicle	16	608.6 $\pm$ 45.5 *	45.6 $\pm$ 4.9 *
	100	12	664.9 $\pm$ 63.9	40.0 $\pm$ 6.8
	300	12	711.5 $\pm$ 89.5	39.6 $\pm$ 7.1
	1000	7	954.2 $\pm$ 82.5#	26.4 $\pm$ 1.4
	3000	8	670.1 $\pm$ 84.2	34.4 $\pm$ 6.0
(b) DKT Normal mice	Vehicle	14	1292.9 $\pm$ 23.3	22.1 $\pm$ 0.9
	100	5	1243.6 $\pm$ 38.0	22.0 $\pm$ 1.2
	300	6	1267.5 $\pm$ 75.8	22.5 $\pm$ 1.1
	1000	7	1293.8 $\pm$ 53.7	20.0 $\pm$ 1.5
	3000	7	1276.4 $\pm$ 96.6	20.0 $\pm$ 0.0

**P* < 0.05 vs normal.

#*P* < 0.05 vs vehicle.

first attenuate gastric motility to determine the efficacy of these substances. Models of attenuated gastric motility induced by clonidine, a central adrenergic α_2 receptor agonist, are currently used to evaluate prokinetic agents.^{4,5} However, our preliminary study showed that we could not reproduce the clonidine-induced gastric motility model in mice (unpublished observation). As mentioned before the paragraph, AITC-induced gastric motility dysfunction might be also attributed to desensitization of parasympathetic nerves by AITC in the gastrointestinal tract, followed by a decrease in ACh release. Therefore, it was thought that AITC administration would create a new model of impaired gastric motility with accompanying mucosal inflammation and attenuated parasympathetic nerve activity.

Interestingly, daikenchuto recovered AITC-induced gastric motility impairment but failed to increase motility under normal conditions. Although it is well known that daikenchuto increases gastrointestinal motility via TRPV1 activation and the subsequent release of ACh and substance P in animal experiments,^{30,31,32} increased gastric motility in normal mice was not observed in this study. These results were consistent with a previous report by Endo et al. showing that daikenchuto increased gastrointestinal motility in a mouse model of postoperative ileus, but not in normal mice.³³ Endo et al. suggested that daikenchuto is more effective at improving motility under pathological conditions compared to that under normal conditions. These findings may explain why daikenchuto increased gastric motility in the AITC-treated mice, but not in untreated mice. We propose three possible reasons for this finding: (1) daikenchuto prevents mucosal inflammation, which is the increase in vascular permeability in response to stimuli (Supplemental Fig. 1); (2) daikenchuto prevents the attenuation of cholinergic activity in response to AITC in the mouse gastrointestinal tract; and (3) components of daikenchuto suppresses TJ barrier disruption in response to inflammation.³⁴

In conclusion, a mouse model of reduced gastric motility was established to evaluate prokinetic agents such as those used to treat FD and traditional Japanese herbal medicines in conscious mice. Oral administration of AITC induced gastric motility dysfunction in mice by disrupting the TJ barrier in the gastric mucosa, increasing vascular permeability, and attenuating cholinergic nerve activity. We propose that this new mouse model be used to help elucidate the pathology of FD.

Declaration of competing interest

The authors have no competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jphs.2020.10.006>.

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